

Simran Christy Serrao

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EDUCATION

Birla Institute of Technology and Science (BITS) Pilani, KK Birla Goa Campus Oct 2022 – Present
B.E. (Hons.) Computer Science; M.Sc. (Hons.) Biological Sciences CGPA: 8.47/10
In-Major GPA (Biological Sciences) : 9.82/10

AWARDS AND FUNDING

- **Khorana Program for Scholars** *UVRI at Stanford Medicine* Jun 2025 - Aug 2025
A collaborative initiative between the Department of Biotechnology (Government of India), WINStep Forward, and the Indo-U.S. Science and Technology Forum that provides Indian students with 10 to 12 weeks of fully funded summer research internships at premier U.S. universities.
- **Merit Scholarship Recipient (Top 3%)** at BITS Pilani KK Birla Goa Campus. Jun 2024

MASTER'S THESIS (ONGOING : JUNE 2026 - JUNE 2027)

An Analysis of Social Network Dynamics among Indian STEM Undergraduates: Implications for Targeted Empathy Interventions

Supervisors: Dr. Ashwini S. , Dr. Indrani Talukdar, Dr. Neena Goveas

- Mapping trust networks among STEM undergraduates to identify central network hubs for targeted empathy interventions.
- Developing a custom Natural Language Processing pipeline for conversational psychometric analysis of Indian STEM cohorts.
- I also apply my work to design real time empathy interventions for the Gender equality cell on campus. I have coordinated and hosted outreach programs including interactive sessions — expression through fine arts, literature, talks, films, dance and music to create awareness regarding gender diversity among undergraduate and graduate students.

RESEARCH EXPERIENCE AND INTERNSHIPS

Stanford Medicine, Department of Pathology Jun 2025 – Sep 2025
Summer Research Intern (Khorana Scholar)
Supervisor: Dr. Le Cong

- Worked on CRISPR-based genome editing in mammalian systems for screening of cancer mutants.
- Data collection and VLM Performance scoring for LabOS:
Cong, L., Smerkous, D., Wang, X., Yin, D., Zhang, Z., Jin, R., Wang, Y., Gerasimiuk, M., Dinesh, R. K., Smerkous, A., Shi, L., Zheng, J., Lam, I., Wu, X., Liu, S., Li, P., Zhu, Y., Zhao, N., Parakh, M., ... **Serrao, S.**, ... Wang, M. (2025). *LabOS: The AI-XR Co-Scientist That Sees and Works With Humans*. arXiv preprint arXiv:2510.14861. <https://arxiv.org/abs/2510.14861>

CSIR-Institute of Microbial Technology, Chandigarh May 2024 – Jul 2024
Summer Research Intern
Supervisor: Dr. Sanjeev Khosla

- Understanding the role of epigenetic modifications in host-pathogen interactions. Isolation of nuclei from Mammalian cells used to conduct MNase sequencing to understand chromatin dynamics and epigenetic regulation.

Unveiling Agency Dynamics of Older Adults in Care Homes in Goa, India, Managed by Churches and Government body Feb 2024 - May 2024
Field Research Assistant *Supervisor: Prof. Nilanjana Goswami*

- Conducted bilingual (English and Konkani) ethnographic interviews with senior citizens in assisted living facilities to gather qualitative data. Coordinated fieldwork operations: the transcription of interview data and the identification of relevant government bodies and NGOs for institutional collaboration.

BITS Pilani, Dept. of Biological Sciences

Aug 2024 – Jan 2025

Supervisor: Dr. Indrani Talukdar

- Analysis of Splice Variant Dataset of hnRNPD: Disease Mapping with Focus on Metabolic, Cardiovascular, and Muscular Disorders.

COURSEWORK AND SKILLS

Programming Languages: Python, Java, C, C++

Humanities Electives: Introductory Psychology, Introduction to Philosophy, Comparative Indian Literature.

Data Science/ML: PyTorch, HuggingFace Transformers, NetworkX, Pandas, NumPy, Scikit-learn

Wet Lab Assays: Mammalian Cell Culture, Nuclei Isolation, MNase-seq Analysis, PCR

Spoken Languages | English, Hindi, Kannada, Konkani, Tulu

LEADERSHIP AND PUBLIC ENGAGEMENT

- President (2024–25) and Secretary (2023–24), Department of Biological Sciences, BITS Goa.
- Student President - Microbiologists Society of India, Goa (2026-2027).
- Member of the Equal Opportunities Committee (2024-2025); Anti-ragging Committee (2023-2024), BITS Pilani, KK Birla, Goa campus.
- Participated in the International Rover Challenge (IRC), 2024 held at PSG iTech, Coimbatore, India, as a member of Project Kratos.
- Volunteer- Teacher at Abhigyaan, BITS Pilani, KK Birla Goa Campus. A student-run organization aimed at educating underprivileged students.
- Core Member, Eco Club, BITS Goa for sustainable ventures.

CERTIFICATIONS & ONLINE COURSEWORK

- **Introduction to Psychology**, Yale University (Coursera) Jan 2024
- **Conspiracy Planet**, UC Santa Cruz (Coursera) Aug 2026
- **Moralities of Everyday Life**, Yale University (Coursera) Aug 2026
- **Mathematics for Machine Learning: Multivariate Calculus**, Imperial College London (Coursera) Sep 2026
- **Soul Beliefs: Causes and Consequences**, Rutgers University (Coursera) Aug 2026
- **Intermediate Visual Communication Skills**, Univ. of Colorado Boulder (Coursera) Aug 2026
- **Narrative Economics**, Yale University (Coursera) Aug 2026
- **Relativism**, University of California, Irvine (Coursera) Sep 2026
- **Social and Economic Networks: Models and Analysis**, Stanford Online (Coursera) Oct 2026

INDEPENDENT PROJECTS

Project Resilience: Neuronal Transition-State Modeling, Independent Research Aug 2026 – Present

- An independent computational systems biology initiative to identify and model multi-omic molecular markers of transition states between neurological compensation and persistent dysfunction.
- We are a team of medical and engineering students, analyzing complex interaction networks to identify dynamical network biomarkers to predict clinical deterioration prior to structural atrophy.

Project Kratos, Life Detection, Mars Rover Team, BITS Goa

Sep 2023 – Sep 2025

- Employing chemical assays and environmental indicators to model the detection of extinct and extant life forms on the Martian surface; understanding the classification of rocks and terrains and their historical potential to support life during their formation process.

MARS Learning Centre, Mangalore Academy of Refined Studies

Jun 2026 – Present

- Digital illustrator for Medical Education